Homework 4

Module 4 - Drug Elimination and Clearance · Homework · No date

Name _____ Date _____

1. Using IBW, estimate the creatinine clearance for a 42-year-old female patient who is 168 cm tall, weighs 63 kg, and has a serum creatinine of 0.88 mg/dL.

Problem 1 IBW

Problem 1 CrClest (First part: IBW = 59.3 kg.)

2. A drug with an elimination half-life of approximately 5 hours and an apparent volume of distribution of 0.25 L/kg was administered to a male volunteer (75 kg) by a

bolus intravenous injection at a dose level of 20 mg/kg.

2a What is the expected plasma concentration 10 hours after administration of the dose?

2b What is the expected AUC in this patient following this IV bolus dose?

2c Urine samples were collected and analyzed for 48 hours following the dose. If 1275 mg of drug was recovered in the urine, what is the excretion rate constant of this agent?

2d What is the renal clearance of this drug?

2e What is the probable mechanism for renal clearance of this drug?

A. Active secretion only
B. Filtration with active secretion
C. Filtration only
D. Filtration with tubular reabsorption

2f What is the hepatic clearance of this drug in this patient?

2g If this drug were administered to a patient with partial renal failure and a clearance of 1.5 L/hr (assuming no change to the volume of distribution), what elimination half-life would you expect?

2h What dose would you recommend to provide the same AUC (as determined in part b) for this patient with partial renal failure and clearance of 1.5L/hr?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

Problem 1 Worked by the drill

Answer: 59.3 kg

WHAT THE STEM GIVES

age	42-year-old	*not needed: IBW depends on height and sex*
height	168 cm	$\frac{168}{2.54}$ = 66.14 in, 6 in over 5 ft
weight	63 kg	*not needed: actual weight is not ideal weight*
S_{Cr}	0.88 mg/dL	*not needed: it enters the CrCl line, not IBW*

THE WORKING, ONE LINE AT A TIME

UNIT

168 cm ÷ 2.54 = 66.14 in → 6 in over 5 ft

The ideal body weight counts whole inches over five feet, so the height is converted and rounded as her keys do.

ALGEBRA

IBW = 45.5 + 2.3(6) = 59.3 kg

The female form of the ideal body weight equation starts at 45.5 kg for a woman 5 ft tall and adds 2.3 kg for each inch over 5 ft: 45.5 + 2.3(6) = 45.5 + 13.8 = 59.3 kg. The 63 kg she weighs is not used, because IBW depends on height and sex alone.

> **DOES THE NUMBER MAKE SENSE?**
> A woman 5 ft tall has an IBW of 45.5 kg, and 168 cm is 6 inches over, so the IBW is 45.5 kg plus 2.3 kg for each of those 6 inches: 59.3 kg, below her actual 63 kg.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem. Her sheet asks this problem as two answer boxes, IBW and CrClest; this one is the IBW box.

Problem 1 Worked by the drill

Answer: 78 mL/min

WHAT THE STEM GIVES

age	42-year-old	140 − 42 = 98 in the numerator
sex	female	multiplies by 0.85
height	168 cm	gave IBW = 59.3 kg in the first part, the numerator
weight	63 kg	*not needed: the question says to use IBW*
S_{Cr}	0.88 mg/dL	the denominator with 72: 72 × 0.88 = 63.36

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{CrCl} = 0.85 \times \frac{(140 - 42)(59.3)}{(72)(0.88)}$$

This is the Cockcroft-Gault equation with ideal body weight; the patient is female, so it is multiplied by 0.85.

ALGEBRA

$$\text{CrCl} = 0.85 \times \frac{5811.4}{63.36} = 78.0 \text{ mL/min}$$

$(140 - 42)(59.3) = 5811.4$ on top and $(72)(0.88) = 63.36$ below; $\frac{5811.4}{63.36} = 91.72$, and $0.85 \times 91.72 = 78.0$ mL/min. The Cockcroft-Gault constants carry the units, so CrCl is reported in mL/min, the unit of the glomerular filtration rate it estimates.

DOES THE NUMBER MAKE SENSE?

A serum creatinine of 0.88 mg/dL is in the normal range for an adult, so CrCl should come out near normal, below 100 mL/min after the 0.85 for a woman: 78.0 mL/min.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem. Her sheet asks this problem as two answer boxes, IBW and CrClest; this one is the CrClest box.

2a

Worked by the drill

Answer: 20.01 mg/L

WHAT THE STEM GIVES

t½	approximately 5 hours	gives $k = \frac{0.693}{5} = 0.1386$ hr^{-1}
V_D	0.25 L/kg	scaled to 75 kg: 18.75 L, the denominator of C_0
weight	75 kg	scales both the dose and the volume
D_0	20 mg/kg	scaled to 75 kg: 1500 mg, the numerator of C_0
t	10 hours after administration	t in e^{-kt}

THE WORKING, ONE LINE AT A TIME

UNIT

$$D_0 = (20 \text{ mg/kg})(75 \text{ kg}) = 1500 \text{ mg}; \; V_D = (0.25 \text{ L/kg})(75 \text{ kg}) = 18.75 \text{ L}; \; k = \frac{0.693}{5} = 0.1386 \text{ hr}^{-1}$$

The dose is 20 mg per kilogram and the volume of distribution V_D is 0.25 L per kilogram, so each is multiplied by the 75 kg weight, the kilograms cancelling: $D_0 = 1500$ mg and $V_D = 18.75$ L. k, the elimination rate constant, is $\frac{0.693}{t½} = \frac{0.693}{5 \text{ hr}} = 0.1386$ hr^{-1}.

ALGEBRA

$$C10 = \frac{1500\ \text{mg}}{18.75\ \text{L}}\ e^{-(0.1386)(10)} = (80)(0.2501) = 20.01\ \text{mg/L}$$

$C_0 = \frac{D_0}{V_D} = \frac{1500\ \text{mg}}{18.75\ \text{L}} = 80$ mg/L is the concentration at time zero; kt = (0.1386)(10) = 1.386, a pure number, and $e^{-1.386} = 0.2501$ is the fraction left after 10 hours, 2.00 half-lives: (80)(0.2501) = 20.01 mg/L.

DOES THE NUMBER MAKE SENSE?

$C_0 = \frac{1500}{18.75} = 80$ mg/L, and 10 hours is two half-lives of 5 hours, so very nearly a quarter of 80 mg/L remains: 20.01 mg/L.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

2b

Worked by the drill

Answer: 577.2 (mg/L)·hr

WHAT THE STEM GIVES

t½	approximately 5 hours	gives k = 0.1386 hr^{-1}, in $Cl_T = kV_D$
V_D	0.25 L/kg	scaled to 75 kg: 18.75 L, in $Cl_T = kV_D = 2.5987$ L/hr
weight	75 kg	scales both the dose and the volume
D_0	20 mg/kg	scaled to 75 kg: 1500 mg, the numerator

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = k \times V_D = (0.1386\ hr^{-1})(18.75\ L) = 2.5987\ L/hr$$

Total clearance Cl_T is the volume of plasma cleared of drug per hour; the fraction k = 0.1386 hr^{-1} of the drug leaves each hour from a volume V_D = 18.75 L, so $Cl_T = k \times V_D = (0.1386\ hr^{-1})(18.75\ L) = 2.5987$ L/hr, reciprocal hours times litres giving litres per hour.

ALGEBRA

$$AUC = \frac{D_0}{Cl_T} = \frac{1500\ \text{mg}}{2.5987\ \text{L/hr}} = 577.2\ \text{(mg/L)·hr}$$

AUC, the area under the concentration–time curve, is the total exposure from the dose; clearance removes drug at $Cl_T \times C$, so over all time $Cl_T \times AUC = D_0$ and $AUC = \frac{D_0}{Cl_T}$.

$\frac{1500\ \text{mg}}{2.5987\ \text{L/hr}}$: dividing by L/hr multiplies by hr/L, so mg becomes (mg/L)·hr, 577.2 (mg/L)·hr.

DOES THE NUMBER MAKE SENSE?

AUC is also $\frac{C_0}{k}$: $C_0 = \frac{1500}{18.75}$ mg/L over 0.1386 hr^{-1} is about 577 (mg/L)·hr, agreeing with 1500 mg over 2.5987 L/hr: 577.2.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

2c

Worked by the drill

Answer: 0.1178 hr^{-1}

WHAT THE STEM GIVES

$t_{1/2}$	approximately 5 hours	gives k = 0.1386 hr^{-1}, multiplied by f_e
V_D	0.25 L/kg	*not needed: f_e and k need no volume*
weight	75 kg	scales the dose to 1500 mg, the denominator of f_e
D_0	20 mg/kg	scaled to 75 kg: 1500 mg, the denominator of f_e
collection	48 hours	*not needed for the number: nearly ten half-lives, so all unchanged drug is in*
D_u	1275 mg	the numerator of f_e

THE WORKING, ONE LINE AT A TIME

SETUP

$$f_e = \frac{1275 \text{ mg}}{1500 \text{ mg}} = 0.85$$

Forty-eight hours is nearly ten half-lives, so the urine holds essentially all the unchanged drug that will be excreted.

ALGEBRA

$$k_e = f_e \times k = (0.85)(0.1386\ hr^{-1}) = 0.1178\ hr^{-1}$$

k = 0.1386 hr^{-1} is the fraction of the drug in the body eliminated per hour by all routes; the kidney accounts for the fraction f_e = 0.85 of that, so the excretion rate constant k_e is that share of k: (0.85)(0.1386 hr^{-1}) = 0.1178 hr^{-1}. f_e has no units, so k_e keeps hr^{-1}.

> **DOES THE NUMBER MAKE SENSE?**
> 1275 of 1500 mg, 0.85 of the dose, left unchanged in urine, so the kidney does 0.85 of the elimination: k_e = 0.85 × 0.1386 = 0.1178 hr^{-1}, below k and above zero.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

2d

Worked by the drill

Answer: 2.209 L/hr

WHAT THE STEM GIVES

$t_{1/2}$	approximately 5 hours	gave $Cl_T = kV_D$ = 2.5987 L/hr in part b
V_D	0.25 L/kg	scaled to 18.75 L, inside Cl_T
weight	75 kg	scales dose and volume
D_0	20 mg/kg	scaled to 1500 mg, the denominator of f_e

collection	48 hours	*not needed for the number: long enough to collect all unchanged drug*
D_u	1275 mg	the numerator of $f_e = 0.85$

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_R = f_e \times Cl_T = (0.85)(2.5987\ \text{L/hr}) = 2.209\ \text{L/hr}$$

Total clearance $Cl_T = 2.5987$ L/hr, from part b, is the plasma volume cleared per hour by all routes. The fraction $f_e = 0.85$ found unchanged in urine is the share the kidney did, so renal clearance $Cl_R = f_e \times Cl_T = (0.85)(2.5987\ \text{L/hr}) = 2.209$ L/hr; f_e has no units.

UNIT

$$2.209\ \text{L/hr} \times \frac{1000\ \text{mL}}{\text{L}} \times \frac{1\ \text{hr}}{60\ \text{min}} = 36.8\ \text{mL/min}$$

L/hr to mL/min: × 1000 mL/L turns litres into millilitres and dividing by 60 min/hr turns hours into minutes, so 2.209 × 1000 = 2209 mL/hr and $\frac{2209\ \text{mL/hr}}{60\ \text{min/hr}} = 36.8$ mL/min. In mL/min the renal clearance can be set beside the glomerular filtration rate, about 120 mL/min, in part e.

ROUND

$$Cl_R = 36.8\ \text{mL/min} \times \frac{1\ \text{L}}{1000\ \text{mL}} \times \frac{60\ \text{min}}{1\ \text{hr}} = 2.208\ \text{L/hr}$$

The question is keyed in L/hr, so the answer is given in L/hr: dividing by 1000 mL/L turns millilitres back into litres and × 60 min/hr turns minutes back into hours, 36.8 × 60 = 2208 mL/hr and $\frac{2208\ \text{mL/hr}}{1000\ \text{mL/L}} = 2.208$ L/hr, the 2.209 L/hr of the first step to rounding.

> **DOES THE NUMBER MAKE SENSE?**
> A fraction 0.85 of the drug leaves by the kidney, so renal clearance is 0.85 of the total 2.5987 L/hr: 2.209 L/hr, below the total; in mL/min that is 36.8.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

2e

Worked by the drill

Answer: D. Filtration with tubular reabsorption

WHAT THE STEM GIVES

t½	approximately 5 hours	gives $k = \frac{0.693}{5} = 0.1386\ \text{hr}^{-1}$, on the way to Cl_T
V_D	0.25 L/kg	scaled to 18.75 L, inside $Cl_T = kV_D$
weight	75 kg	scales the dose and the volume
D_0	20 mg/kg	scaled to 1500 mg, the denominator of f_e
collection	48 hours	*not needed for the number: long enough to collect all unchanged drug*
D_u	1275 mg	the numerator of $f_e = 0.85$

THE WORKING, ONE LINE AT A TIME

SETUP

$D_0 = (20\ \text{mg/kg})(75\ \text{kg}) = 1500\ \text{mg}$; $f_e = \dfrac{D_u}{D_0} = \dfrac{1275\ \text{mg}}{1500\ \text{mg}} = 0.85$

f_e is the fraction of the dose reaching the urine unchanged. It says how much of the elimination is renal, 85%, not how the kidney does it.

SETUP

$k = \dfrac{0.693}{5} = 0.1386\ \text{hr}^{-1}$; $V_D = (0.25\ \text{L/kg})(75\ \text{kg}) = 18.75\ \text{L}$; $Cl_T = kV_D = (0.1386)(18.75) = 2.5987\ \text{L/hr}$

The mechanism is read from renal clearance, a share of total clearance, so total clearance comes first: k from the half-life and the volume scaled to the patient.

ALGEBRA

$Cl_R = f_e \times Cl_T = 0.85 \times 2.5987\ \text{L/hr} = 2.209\ \text{L/hr}$

Renal clearance is the renal fraction of total clearance: f_e now scales a clearance, which has units and a physiological reference to compare against.

UNIT

$Cl_R = 2.209\ \text{L/hr} \times \dfrac{1000\ \text{mL}}{1\ \text{L}} \times \dfrac{1\ \text{hr}}{60\ \text{min}} = 36.8\ \text{mL/min}$

The glomerular filtration rate is quoted in mL/min, so renal clearance is put in the same unit before the two are compared.

SETUP

Compare: $Cl_R = 36.8$ mL/min against GFR of about 120 mL/min; 36.8 is below 120

A renal clearance below the GFR means some filtered drug is taken back into the blood: filtration with partial reabsorption.

> **DOES THE NUMBER MAKE SENSE?**
> $f_e = 0.85$ alone cannot name the mechanism: a drug with the same f_e but a much shorter half-life would have a renal clearance above GFR and be secreted. Only Cl_R in mL/min set against about 120 mL/min decides, and 36.8 is well below it.

Renal clearance is 36.8 mL/min, below the GFR, the glomerular filtration rate of about 120 mL/min. Filtration alone would clear plasma at that rate, so some filtered drug moved back into the blood: filtration with partial reabsorption.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

2f Worked by the drill

Answer: 0.39 L/hr

WHAT THE STEM GIVES

t½	approximately 5 hours	gave $Cl_T = 2.5987$ L/hr in part b, the total
V_D	0.25 L/kg	inside Cl_T

weight	75 kg	scales dose and volume
D_0	20 mg/kg	the denominator of f_e
collection	48 hours	*not needed for the number*
D_u	1275 mg	gave $f_e = 0.85$ and $Cl_R = 2.209$ L/hr in part d, subtracted

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_H = Cl_T - Cl_R = 2.5987 - 2.209 = 0.390 \text{ L/hr}$$

Total clearance Cl_T is the sum of the clearances by each route, $Cl_T = Cl_R + Cl_H$, so the hepatic clearance Cl_H is what remains after the renal clearance Cl_R from part d is subtracted: $2.5987 - 2.209 = 0.390$ L/hr, the fraction $1 - f_e = 0.15$ of the total.

> **DOES THE NUMBER MAKE SENSE?**
> The kidney does most of the elimination, so the liver does the small remainder of 2.5987 L/hr: 0.390 L/hr, which is also 2.5987 − 2.209; it is below the renal clearance and above zero.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

2g

Worked by the drill

Answer: 8.66 hr

WHAT THE STEM GIVES

t½	approximately 5 hours	the normal half-life, replaced by the new one; not in the line
V_D	0.25 L/kg	scaled to 18.75 L, unchanged, the numerator with 0.693
weight	75 kg	scales the volume to 18.75 L
D_0	20 mg/kg	*not needed: half-life does not depend on the dose*
new Cl	1.5 L/hr	the denominator

THE WORKING, ONE LINE AT A TIME

SETUP

$$t½ = \frac{0.693\ V_D}{Cl} = \frac{0.693(18.75\text{ L})}{1.5\text{ L/hr}} = 8.66 \text{ hr}$$

$t½ = \frac{0.693\ V_D}{Cl}$ combines $Cl = k \times V_D$ with $k = \frac{0.693}{t½}$. V_D is unchanged at 18.75 L, so $(0.693)(18.75\text{ L}) = 12.994$ L and $\frac{12.994\text{ L}}{1.5\text{ L/hr}} = 8.66$ hr. Clearance fell from 2.5987 to 1.5 L/hr, so the half-life rose from 5 hr.

> **DOES THE NUMBER MAKE SENSE?**
> Clearance fell to 1.5 L/hr in partial renal failure with the volume unchanged, so the half-life rises above the normal 5 hr: 8.66 hr.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

2h

Worked by the drill

Answer: 865.8 mg

WHAT THE STEM GIVES

t½	approximately 5 hours	gave the part b AUC of 577.2 (mg/L)·hr
V_D	0.25 L/kg	inside the part b AUC
weight	75 kg	scales the dose to 1500 mg, the normal dose
D_0	20 mg/kg	1500 mg gave the AUC at normal clearance; the new dose is asked
new Cl	1.5 L/hr	multiplied by the AUC

THE WORKING, ONE LINE AT A TIME

SETUP

$$D_0 = AUC \times Cl = (577.2\ (mg/L)\cdot hr)(1.5\ L/hr) = 865.8\ mg$$

$AUC = \frac{D_0}{Cl}$ rearranged: a lower clearance needs a proportionally lower dose to give the same exposure.

> **DOES THE NUMBER MAKE SENSE?**
>
> Dose and clearance scale together for the same AUC: clearance fell to 1.5 L/hr, a little over half of normal, so the dose falls to a little over half of the normal 20 mg/kg × 75 kg: 865.8 mg, below it.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.
